

EP-based Distributed Message Passing Detection for Cell-Free Massive MIMO

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Abstract—Expectation propagation (EP) has been applied to signal detection in Cell-Free massive MIMO (CF-mMIMO) systems to achieve high performance. However, its computational complexity becomes prohibitive as the number of users and antennas increases. In this letter, we apply the EP-based approximate message passing framework to CF-mMIMO and propose a distributed detector, called EP-dAMP, which efficiently distributes computational tasks between the access points (APs) and the central processing unit (CPU) by offloading part of the computations from the APs to the CPU. Building on this foundation, we further propose a low-complexity variant, LC-dAMP, designed to reduce the computational demands on APs and achieve lower fronthaul overhead. Simulation results demonstrate that both algorithms can achieve a complexity reduction of 85%-94% compared to existing non-linear detectors for CF-mMIMO while achieving better bit error rate (BER) performance.

Index Terms—Cell-Free massive MIMO, expectation propagation, approximate message passing, fronthaul overhead.

I. INTRODUCTION

TO ADDRESS the inter-cell interference problem in conventional cellular networks, Cell-Free massive multiple-input multiple-output (CF-mMIMO) has emerged as a promising technology for next-generation wireless communication systems [1]. Unlike traditional co-located MIMO systems that rely on fixed base stations (BSs) serving users within predefined cells, CF-mMIMO employs a distributed architecture where numerous access points (APs) are densely deployed across the coverage area. These APs are connected to a central processing unit (CPU) via high-speed fronthaul links, enabling coordinated signal processing [2]. By placing APs closer to edge users, CF-mMIMO mitigates inter-cell interference, thereby improving the spectral efficiency (SE) [3].

Efficient detection algorithms are crucial for maximizing the benefits of CF-mMIMO. Conventional detectors rely on maximum ratio combining (MRC) [1] at each AP for local data detection. However, according to [4], [5], advanced detection methods have been shown to achieve greater SE than simpler, MRC-based detection. In particular, four distinct detection schemes for CF-mMIMO, representing varying

degrees of AP/CPU cooperation, were analyzed in [4]. Among these, the fully-centralized linear minimum mean square error (LMMSE) algorithm achieves the highest SE [4]. Nevertheless, as a linear receiver, the bit-error rate (BER) performance of LMMSE remains suboptimal. A centralized Variational Bayesian (ceVB) algorithm was proposed in [6], which showed superior BER performance than LMMSE under 4-QAM. However, its prohibitively high complexity poses significant challenges for implementation.

To address this limitation, recent research has explored non-linear distributed algorithms. For example, a decentralized expectation propagation (EP)-based detector was developed in [7] for channel-coded uplink mMIMO receivers. Furthermore, an iterative detection algorithm for CF-mMIMO systems, proposed in [8], leveraged the distributed EP principle [9], integrating a linear MMSE module at each AP and a non-linear module at the CPU for centralized data processing. Similarly, [10] investigated a joint estimation and detection scheme based on the bilinear EP algorithm. However, the decentralized EP [7] cannot be directly applied to CF-mMIMO. Its localized iterative approach poses significant convergence challenges when the number of receive antennas at each AP is smaller than the number of users, a common condition in CF-mMIMO. Furthermore, as the number of users and antennas increases, the high dimensional matrix operations in distributed EP [8] and bilinear EP [10] would result in prohibitively high computational complexity at each AP. In addition, bilinear EP also suffers from computation precision issues.

In this letter, we introduce an EP-based distributed approximate message passing (EP-dAMP) algorithm for uplink detection in CF-mMIMO systems. To further reduce computational complexity, we then propose a low-complexity distributed approximate message passing (LC-dAMP) detection algorithm. The main contributions of this letter are as follows: (1) Based on the fully factorized factor graph, EP-dAMP is efficiently developed for CF-mMIMO, with its computational load strategically distributed between the APs and the CPU. (2) By leveraging insights from the antenna ratio in CF-mMIMO, LC-dAMP reformulates the message passing rules, reducing the computational resources required at the APs, while maintaining performance similar to EP-dAMP. (3) Simulation results show that the proposed algorithms outperform existing centralized and distributed detectors for CF-mMIMO while maintaining low computational complexity.

II. PRELIMINARIES

A. CF-mMIMO System Model

Consider an uplink CF-mMIMO system consisting of L APs and K single-antenna users. Each AP is equipped with M

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antennas and connected to a highly capable CPU via fronthaul links. With CPU coordination, all APs can concurrently provide service to all users in the same time-frequency resource. During the uplink data transmission phase, all K users transmit information to the APs simultaneously. In each coherence block, the uplink communication comprises τ_p mutually orthogonal pilot signals for channel estimation and $\tau_c - \tau_p$ channel uses for data transmission. The channel between the k -th user and the l -th AP is modeled as $\mathbf{h}_{l,k} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, \mathbf{R}_{l,k})$, where $\mathbf{R}_{l,k} \in \mathbb{C}^{M \times M}$ denotes the spatial correlation matrix. The received baseband signal $\mathbf{y}_l = [y_{l,1}, \dots, y_{l,M}]^T \in \mathbb{C}^{M \times 1}$ at the l -th AP is obtained by

$$\mathbf{y}_l = \sum_k \sqrt{p_k} \mathbf{h}_{l,k} x_k + \mathbf{n}_l, \quad (1)$$

where $p_k > 0$ is the transmit power of the k -th user, $x_k \in \Theta$ denotes the transmitted information-bearing symbol mapped from the Q -QAM constellation Θ , and $\mathbf{n}_l \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, \sigma_n^2 \mathbf{I}_M)$ is the additive white Gaussian noise (AWGN) at the receiver side. In a more compact form, we denote channel matrix between the l -th AP and all users as $\mathbf{H}_l = [\mathbf{h}_{l,1}, \dots, \mathbf{h}_{l,K}] \in \mathbb{C}^{M \times K}$. The uplink data detection task for CF-mMIMO is to infer the transmitted signals $\mathbf{x} = [x_1, \dots, x_K]^T$ given the channel matrix $\mathbf{H} = [\mathbf{H}_1^T, \dots, \mathbf{H}_L^T]^T \in \mathbb{C}^{LM \times K}$ between all users and all APs, received signals $\mathbf{y} = [\mathbf{y}_1^T, \dots, \mathbf{y}_L^T]^T \in \mathbb{C}^{LM \times 1}$, and noise variance σ_n^2 . The posterior distribution of $\mathbf{x}$ is given by

$$\begin{aligned} p(\mathbf{x}|\mathbf{y}, \mathbf{H}) &= p_r(\mathbf{x}) \prod_l p(\mathbf{y}_l|\mathbf{x}, \mathbf{H}_l) \\ &\propto p_r(\mathbf{x}) \prod_l \mathcal{N}_{\mathbb{C}}(\mathbf{y}_l; \mathbf{H}_l \mathbf{x}, \sigma_n^2 \mathbf{I}_M), \end{aligned} \quad (2)$$

where $p_r(\mathbf{x}) = \frac{1}{|\Theta|^K} \prod_k \mathbb{I}_{x_k \in \Theta}$ is the joint prior distribution of $\mathbf{x}$ and $p(\mathbf{y}_l|\mathbf{x}, \mathbf{H}_l)$ is the l -th likelihood probability.

B. EP-Based Approximate Message Passing Detectors

As a specific application of the sum-product algorithm, the message passing detector uses the distributive law to calculate the marginal function of the joint distribution $p(\mathbf{x}, \mathbf{y}_l|\mathbf{H})$, which can be factorized as

$$p(\mathbf{x}, \mathbf{y}_l|\mathbf{H}) \propto p_r(\mathbf{x}) \prod_m \exp \left\{ -\frac{|y_{l,m} - \sum_k h_{l,mk} x_k|^2}{\sigma_n^2} \right\}, \quad (3)$$

where $h_{l,mk}$ is the m -th entry of the channel vector $\mathbf{h}_{l,k}$.

Eq. (3) can be depicted by the factor graph (FG) shown in Fig. 1. Define $\phi_{x_k \rightarrow f_{l,m}}(x_k)$ and $\phi_{f_{l,m} \rightarrow x_k}(x_k)$ as the beliefs passed from the variable node (VN) x_k to the FN $f_{l,m}$ and vice versa, respectively. The update rule in FG is derived as follows [11],

$$\phi_{x_k \rightarrow f_{l,m}}(x_k) = \phi_{\theta_k \rightarrow x_k}(x_k) \prod_{m' \neq m} \phi_{f_{l,m'} \rightarrow x_k}(x_k), \quad (4a)$$

$$\phi_{f_{l,m} \rightarrow x_k}(x_k) = \sum_{\mathbf{x} \setminus x_k} f_{l,m}(y_{l,m}|\mathbf{x}) \prod_{k' \neq k} \phi_{x_{k'} \rightarrow f_{l,m}}(x_{k'}), \quad (4b)$$

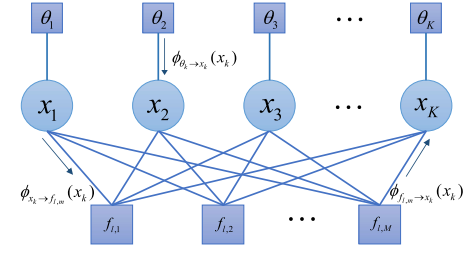


Fig. 1. Factor graph representation of CF-mMIMO system for one AP.

where $\phi_{\theta_k \rightarrow x_k}(x_k)$ denotes the prior information passed from the mapping node θ_k to x_k . The marginal posterior distribution of the k -th symbol $p(x_k)$ can then be given as

$$p(x_k) \propto \phi_{\theta_k \rightarrow x_k}(x_k) \prod_m \phi_{f_{l,m} \rightarrow x_k}(x_k). \quad (5)$$

The basic idea of EP-based AMP is to eliminate the exhaustive traversal of all constellation points for each passed message in FG by using EP technique [12]. Specifically, the posterior probability $p(x_k)$ and belief $\phi_{x_k \rightarrow f_{l,m}}(x_k)$ are approximated as complex Gaussian distributions $\mathcal{N}_{\mathbb{C}}(\hat{x}_k, \hat{v}_k)$ and $\mathcal{N}_{\mathbb{C}}(x_{k'} \rightarrow f_{l,m}, v_{x_{k'} \rightarrow f_{l,m}})$, respectively. By moment matching condition [13], $\hat{x}_k$ and $\hat{v}_k$ are obtained by

$$\hat{x}_k = \mathbb{E}_{p(x_k)}(x_k), \quad \hat{v}_k = \text{Var}_{p(x_k)}(x_k). \quad (6)$$

By this approximation, $\phi_{f_{l,m} \rightarrow x_k}(x_k)$ also becomes a complex Gaussian distribution denoted by $\mathcal{N}_{\mathbb{C}}(x_{f_{l,m} \rightarrow x_k}, v_{f_{l,m} \rightarrow x_k})$, as $\phi_{f_{l,m} \rightarrow x_k} \propto \frac{p(x_k)}{\phi_{x_k \rightarrow f_{l,m}}}$. Eq. (4b) can then be rewritten in the integration form

$$\phi_{f_{l,m} \rightarrow x_k}(x_k) = \int_{\mathbf{x} \setminus x_k} f_{l,m}(y_{l,m}|\mathbf{x}) \prod_{k' \neq k} \phi_{x_{k'} \rightarrow f_{l,m}}(x_{k'}). \quad (7)$$

Based on Eqs. (4a) and (7), the beliefs traveling back and forth in FG can be calculated as follows:

$$v_{f_{l,m} \rightarrow x_k} = \frac{\sum_{k' \neq k} |h_{l,mk'}|^2 v_{x_{k'} \rightarrow f_{l,m}} + \sigma_n^2}{|h_{l,mk}|^2}, \quad (8a)$$

$$x_{f_{l,m} \rightarrow x_k} = \frac{y_{l,m} - \sum_{k' \neq k} h_{l,mk'} x_{x_{k'} \rightarrow f_{l,m}}}{h_{l,mk}}, \quad (8b)$$

$$v_{x_k \rightarrow f_{l,m}} = \left(\frac{1}{\hat{v}_k} - \frac{1}{v_{f_{l,m} \rightarrow x_k}} \right)^{-1}, \quad (8c)$$

$$x_{x_k \rightarrow f_{l,m}} = v_{x_k \rightarrow f_{l,m}} \left(\frac{\hat{x}_k}{\hat{v}_k} - \frac{x_{f_{l,m} \rightarrow x_k}}{v_{f_{l,m} \rightarrow x_k}} \right). \quad (8d)$$

III. THE PROPOSED DISTRIBUTED APPROXIMATE MESSAGE PASSING DETECTORS

The limited computational resources at each AP pose significant challenges for conventional distributed EP detectors [8], [9], particularly as the number of users increases, making high-order matrix operations increasingly expensive. In this section, we present the EP-dAMP detector for CF-mMIMO, built on the EP-based AMP algorithm [14], and designed to effectively allocate computational overhead between the CPU and the APs. Additionally, we introduce

Algorithm 1: EP-dAMP for CF-mMIMO

Input : $\mathbf{y}_l, \mathbf{H}_l, \sigma_n^2$
Output: $\mathbf{x}^{ext, T}, \mathbf{v}^{ext, T}$

- 1 Initialization: $x_{k \rightarrow f_{l,m}}^0 = 0, v_{x_k \rightarrow f_{l,m}}^0 = 1$.
- 2 For each iteration $t = 1, 2, \dots, T$:
- 3 **Operations in APs:**
- 4 **for** $m = 1, 2, \dots, M$ **do**
- 5 **for** $k = 1, 2, \dots, K$ **do**
- 6 $x_{f_{l,m} \rightarrow x_k}^t = \frac{z_{f_{l,m}}^t + h_{l,mk} x_{x_k \rightarrow f_{l,m}}^t}{h_{l,mk}},$
- 7 $v_{f_{l,m} \rightarrow x_k}^t = \frac{\tau_{f_{l,m}}^t - |h_{l,mk}|^2 v_{x_k \rightarrow f_{l,m}}^t}{|h_{l,mk}|^2}.$
- 8 **Operations in the CPU:**
- 9 **for** $k = 1, 2, \dots, K$ **do**
- 10 $v_k^{ext, t} = \left(\sum_l \sum_m \frac{1}{v_{f_{l,m} \rightarrow x_k}^t} \right)^{-1},$
- 11 $x_k^{ext, t} = v_k^{ext, t} \left(\sum_l \sum_m \frac{x_{f_{l,m} \rightarrow x_k}^t}{v_{f_{l,m} \rightarrow x_k}^t} \right).$
- 12 **if** $t < T$ **then**
- 13 **for** $k = 1, 2, \dots, K$ **do**
- 14 $\hat{p}^t(x_k) = \frac{1}{Z} \phi_{\theta_k \rightarrow x_k}^t(x_k) \mathcal{N}_C(x_k^{ext, t}, v_k^{ext, t})$
- 15 $\hat{x}_k^t = \mathbb{E}_{\hat{p}^t(x_k)}(x_k), \hat{v}_k^t = \text{Var}_{\hat{p}^t(x_k)}(x_k).$
- 16 **for** $m = 1, 2, \dots, M$ **do**
- 17 $v_{x_k \rightarrow f_{l,m}}^{t+1} = \left(\frac{1}{\hat{v}_k^t} - \frac{1}{v_{f_{l,m} \rightarrow x_k}^t} \right)^{-1},$
- 18 $x_{x_k \rightarrow f_{l,m}}^{t+1} = v_{x_k \rightarrow f_{l,m}}^{t+1} \left(\frac{\hat{x}_k^t}{\hat{v}_k^t} - \frac{x_{f_{l,m} \rightarrow x_k}^t}{v_{f_{l,m} \rightarrow x_k}^t} \right).$

a low-complexity variant of EP-dAMP, called LC-dAMP, designed to reduce both overall complexity and fronthaul overhead.

A. EP-Based Distributed AMP

First, by distributing EP-based AMP in the CF-mMIMO framework, we propose the EP-dAMP detector as summarized in Algorithm 1. With $\mathbf{y}_l, \mathbf{H}_l$, and noise variance σ_n^2 available at l -th AP, the t -th EP-dAMP iteration at each AP calculates the Gaussian parameters $x_{f_{l,m} \rightarrow x_k}^t$ and $v_{f_{l,m} \rightarrow x_k}^t$ based on Eqs. (8a) and (8b). Note that instead of tracking $MK(K-1)$ messages to update $x_{f_{l,m} \rightarrow x_k}^t$ and $v_{f_{l,m} \rightarrow x_k}^t$ at each AP, we can reduce the number of tracked messages to MK by introducing the aggregate parameters $z_{f_{l,m}}^t = y_{l,m} - \sum_k h_{l,mk} x_{x_k \rightarrow f_{l,m}}^t$ and $\tau_{f_{l,m}}^t = \sum_k |h_{l,mk}|^2 v_{x_k \rightarrow f_{l,m}}^t + \sigma_n^2$. Then Eqs. (8a) and (8b) are reformulated as

$$x_{f_{l,m} \rightarrow x_k}^t = \frac{z_{f_{l,m}}^t + h_{l,mk} x_{x_k \rightarrow f_{l,m}}^t}{h_{l,mk}}, \quad (9a)$$

$$v_{f_{l,m} \rightarrow x_k}^t = \frac{\tau_{f_{l,m}}^t - |h_{l,mk}|^2 v_{x_k \rightarrow f_{l,m}}^t}{|h_{l,mk}|^2}, \quad (9b)$$

After all the beliefs of the L APs are updated, the CPU gathers messages from each AP to renew the extrinsic information

of $\prod_l p(\mathbf{y}_l | \mathbf{x}, \mathbf{H}_l)$. As a product of Gaussians, the variance and mean can be calculated by

$$v_k^{ext, t} = \left(\sum_l \sum_m \frac{1}{v_{f_{l,m} \rightarrow x_k}^t} \right)^{-1}, \quad (10a)$$

$$x_k^{ext, t} = v_k^{ext, t} \left(\sum_l \sum_m \frac{x_{f_{l,m} \rightarrow x_k}^t}{v_{f_{l,m} \rightarrow x_k}^t} \right). \quad (10b)$$

Based on the obtained $x_k^{ext, t}$, $v_k^{ext, t}$, and prior probability $\phi_{\theta_k \rightarrow x_k}^t(x_k)$, the approximate posterior probability of the k -th symbol is given by

$$\hat{p}^t(x_k) = \frac{1}{Z} \phi_{\theta_k \rightarrow x_k}^t(x_k) \mathcal{N}_C(x_k^{ext, t}, v_k^{ext, t}), \quad (11)$$

where Z denotes the normalization factor. Then, for each user, we traverse M receive antennas and compute the messages propagated from VNs to FNs using Eqs. (6), (8c), and (8d). When t -th iteration is completed at this point, $v_{x_k \rightarrow f_{l,m}}^t$ and $x_{x_k \rightarrow f_{l,m}}^t$ are fed back from the CPU to each AP for the next loop until the maximum of iterations T is reached.

B. Low-Complexity Distributed AMP

In the proposed EP-dAMP, the computational complexity of Eq. (9) remains prohibitive for each AP. By leveraging the nature of the CF-mMIMO antenna configuration, we propose an LC-dAMP detector that reformulates the message passing rules in EP-dAMP, further reducing the AP-side complexity. According to Eq. (10a), the variance $\hat{v}_k^t$ is approximately $1/ML$ of the harmonic mean of all $v_{f_{l,m} \rightarrow x_k}^t$. Given that $ML \gg 1$ in CF networks, the average amplitude of $v_{f_{l,m} \rightarrow x_k}^t$ is much larger than $\hat{v}_k^t$. As a result, $v_{x_k \rightarrow f_{l,m}}^{t+1}$ can be approximated as $\hat{v}_k^t$. Bringing $v_{x_k \rightarrow f_{l,m}}^{t+1} = \hat{v}_k^t$ into Eq. (8d), $x_{x_k \rightarrow f_{l,m}}^{t+1}$ is reformulated as

$$x_{x_k \rightarrow f_{l,m}}^{t+1} = \hat{x}_k^t - \frac{\hat{v}_k^t x_{f_{l,m} \rightarrow x_k}^t}{v_{f_{l,m} \rightarrow x_k}^t}, \quad (12)$$

where the absolute value of residual term $\left| \frac{v_k^t x_{f_{l,m} \rightarrow x_k}^t}{v_{f_{l,m} \rightarrow x_k}^t} \right| < \frac{v_k^t}{v_{f_{l,m} \rightarrow x_k}^t}$ can be regarded as $\mathcal{O}(\frac{1}{ML})$, and can thus be safely neglected. Given the approximations of messages propagated from VNs to FNs, $z_{f_{l,m}}^t$ and $\tau_{f_{l,m}}^t$ can be rewritten as

$$z_{f_{l,m}}^t = y_{l,m} - \sum_k h_{l,mk} \hat{x}_k^t, \quad (13a)$$

$$\tau_{f_{l,m}}^t = \sum_k |h_{l,mk}|^2 \hat{v}_k^t + \sigma_n^2. \quad (13b)$$

According to [12], the term $|h_{l,mk}|^2 \hat{v}_k^t$ may be omitted in Eq. (9b) if K is large. With $v_{f_{l,m} \rightarrow x_k}^t = \tau_{f_{l,m}}^t / |h_{l,mk}|^2$, the extrinsic mean of each symbol can be calculated as

$$x_k^{ext, t} = v_k^{ext, t} \left(\sum_l \sum_m \frac{h_{l,mk}^* z_{f_{l,m}}^t + |h_{l,mk}|^2 \hat{x}_k^t}{\tau_{f_{l,m}}^t} \right)$$

Algorithm 2: LC-dAMP for CF-mMIMO

Input : $\mathbf{y}_l, \mathbf{H}_l, \sigma_n^2$
Output: $\mathbf{x}^{ext,T}, \mathbf{v}^{ext,T}$

- 1 Initialization: $\hat{x}_k^0 = 0, \hat{v}_k^0 = 1$.
- 2 For each iteration $t = 1, 2, \dots, T$:
- 3 **Operations in APs:**
- 4 **for** $m = 1, 2, \dots, M$ **do**
- 5 $z_{f_l,m}^t = y_{l,m} - \sum_k h_{l,mk} \hat{x}_k^t$,
- 6 $\tau_{f_l,m}^t = \sum_k |h_{l,mk}|^2 \hat{v}_k^t + \sigma_n^2$.
- 7 **Operations in the CPU:**
- 8 **for** $k = 1, 2, \dots, K$ **do**
- 9 $v_k^{ext,t} = \left(\sum_l \sum_m \frac{|h_{l,mk}|^2}{\tau_{f_l,m}^t} \right)^{-1}$,
- 10 $x_k^{ext,t} = \hat{x}_k^t + v_k^{ext,t} \left(\sum_l \sum_m \frac{h_{l,mk}^* z_{f_l,m}^t}{\tau_{f_l,m}^t} \right)$.
- 11 **if** $t < T$ **then**
- 12 $\hat{p}^t(x_k) = \frac{1}{Z} \phi_{\theta_k \rightarrow x_k}^t(x_k) \mathcal{N}_{\mathbb{C}}(x_k^{ext,t}, v_k^{ext,t})$,
- 13 $\hat{x}_k^{t+1} = \mathbb{E}_{\hat{p}^t(x_k)}(x_k), \hat{v}_k^{t+1} = \mathbb{V}ar_{\hat{p}^t(x_k)}(x_k)$.

TABLE I
FRONTAUL OVERHEAD PER CHANNEL USE

Algorithms	Each coherence block	Statistical parameters
LMMSE/ceVB	$\tau_c ML$	$M^2 KL/2$
deEP	$2(\tau_c - \tau_p)(K+1)LT$	—
EP-dAMP	$2(\tau_c - \tau_p)MKLT$	—
LC-dAMP	$2(\tau_c - \tau_p)MLT$	$2MKL$

$$= \hat{x}_k^t + v_k^{ext,t} \left(\sum_l \sum_m \frac{h_{l,mk}^* z_{f_l,m}^t}{\tau_{f_l,m}^t} \right). \quad (14)$$

Furthermore, EP-dAMP suffers from relatively high fronthaul overhead as there are $2(\tau_c - \tau_p)MKLT$ complex parameters to be transmitted from the APs to the CPU altogether. To reduce the fronthaul overhead of EP-dAMP in LC-dAMP, the channel statistical parameters $h_{l,mk}$ and $|h_{l,mk}|^2$ are first passed from the APs to the CPU during each coherence block. $v_k^{ext,t}$ is therefore rewritten as

$$v_k^{ext,t} = \left(\sum_l \sum_m \frac{|h_{l,mk}|^2}{\tau_{f_l,m}^t} \right)^{-1}. \quad (15)$$

Consequently, the overall overhead for LC-dAMP is $2(\tau_c - \tau_p)MLT$ complex scalars, accompanied by $2MKL$ complex statistical parameters. The proposed LC-dAMP is summarized in Algorithm 2. The fronthaul overhead of the proposed detectors is compared with centralized LMMSE/ceVB and distributed EP (deEP) in terms of complex scalars in Table I.

IV. SIMULATION RESULTS

In this section, we compare the numerical results of the proposed EP-dAMP and LC-dAMP with those of deEP, centralized LMMSE, and ceVB under various scenarios.

In Fig. 2 with $L = 16, K = 16, M = 8$, the shadowing fading coefficients are set to unit with the large-scale fading

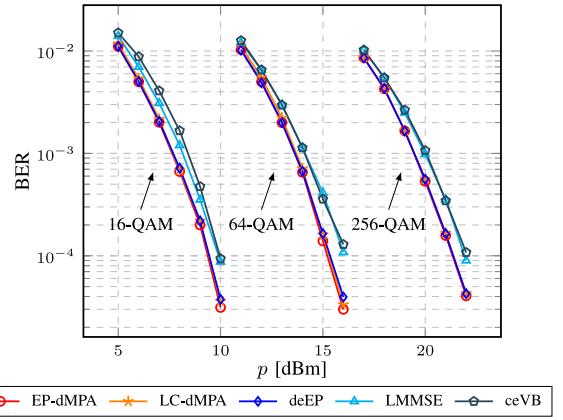


Fig. 2. BER performance of the proposed EP-dAMP and LC-dAMP compared to deEP [8], centralized LMMSE [5] and ceVB [6] at different modulation orders. The configuration $L \times K \times M = 16 \times 16 \times 8$.

TABLE II
NUMBER OF REAL-VALUED MULTIPLICATIONS OF DIFFERENT DETECTORS

Module	Algorithms	NRMs
AP	LMMSE	—
	ceVB	—
	deEP	$(4MK + 16M^2K + 8MK^2 + 16M^3/3)LT$
	EP-dAMP	$4MKL + 24MKLT$
	LC-dAMP	$4MKL + 8MKLT$
CPU	LMMSE	$8MK^2L + 16K^3/3 + 4K^2$
	ceVB	$8MK^2L + 29K^3T + 16K^2T + 14\sqrt{Q}KT$
	deEP	$(8KL + 14\sqrt{Q}K + 4L + 4K)T$
	EP-dAMP	$(16MKL + 14\sqrt{Q}K + 8K)T$
	LC-dAMP	$(8MKL + 14\sqrt{Q}K + 4K + 2ML)T$

coefficients fixed. It can be observed that both EP-dAMP and LC-dAMP exhibit a performance improvement of about 0.5 dB over LMMSE and ceVB at the BER of 1×10^{-4} for all modulation orders. Additionally, the proposed algorithms achieve similar BER performance to deEP but with significantly lower computational complexity, which is verified in the next section.

In Fig. 3, we consider a more practical scenario which accounts for shadowing fading and large-scale fading variations, aligning the channel parameters with those used in [4]. Inspired by [15], we evaluate the proposed algorithms given imperfect channel state information (CSI). In Fig. 3(b), we assume the CF system experiences pilot contamination, with the number of mutually orthogonal pilots set to $\tau_p = 30 < K$. The results demonstrate that both EP-dAMP and LC-dAMP outperform the ceVB method, with a performance improvement of approximately 1.2 dB at higher transmit power regime under both perfect and imperfect CSI conditions. The LMMSE method exhibits an approximate loss of 2.4 dB compared to the proposed algorithms, while the performance of deEP shows significant degradation in both cases.

V. COMPLEXITY ANALYSIS

The number of real-valued multiplications (NRMs) for the proposed EP-dAMP and LC-dAMP is compared with other detectors in Table II. Due to the limitation of table

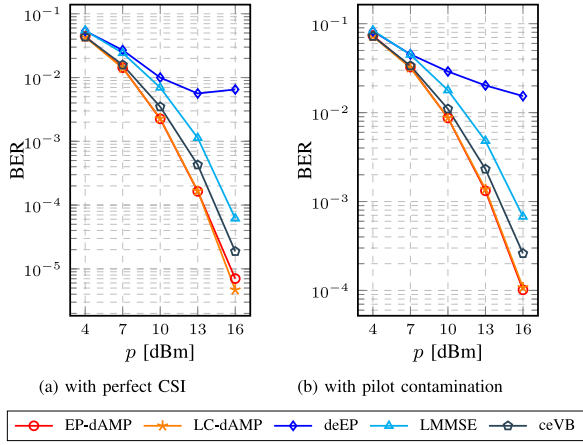


Fig. 3. The BER performance of EP-dAMP and LC-dAMP compared to deEP, centralized LMMSE and ceVB given perfect/imperfect CSI with $L = 32$, $K = 32$, and $M = 4$ under 4-QAM.

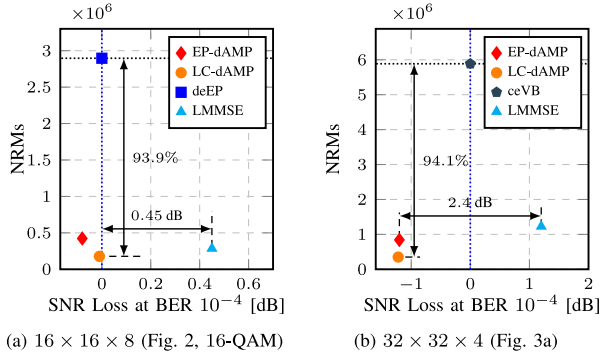


Fig. 4. Comparison of the trade-off between complexity and SNR loss at $\text{BER} = 10^{-4}$ under the scenarios in Fig. 2 (16-QAM) and in Fig. 3(a).

space, some low-order terms of operation number are omitted. As shown in Eqs. (9) and (13), EP-dAMP and LC-dAMP require only $\mathcal{O}(MKT + MK)$ operations for message passing from FNs to VNs at each AP. As listed in Table II, with $T = 5$, LC-dAMP reduces the AP-side computational cost by approximately 60% compared to the EP-dAMP detector. Meanwhile, the computational complexity of EP-dAMP and LC-dAMP at the CPU side mainly consists of message fusion and moment matching operations, which are at the order of $\mathcal{O}(MKLT + \sqrt{Q}KT)$. As detailed in Table II, EP-dAMP and LC-dAMP exhibit comparable NRMs with deEP in terms of CPU processing, while significantly reducing complexity in each AP. Notably, the computational complexity of ceVB is the highest among all methods evaluated.

Fig. 4 further illustrates the trade-off between complexity and performance for the proposed detectors in the scenarios depicted in Fig. 2 (16-QAM) and Fig. 3(a). The horizontal axis represents the SNR loss of different algorithms compared at $\text{BER} = 10^{-4}$, while the vertical axis indicates the NRMs of each detectors. As shown in Fig. 4(a), EP-dAMP reduces

the complexity of deEP by 85% while demonstrating better performance. LC-dAMP, on the other hand, conserves 93.9% of the computational burden of deEP with similar performance. In Fig. 4(b), EP-dAMP achieves a 2.4 dB gain over LMMSE with smaller NRMs. Additionally, LC-dAMP reduces the complexity to approximately 5.9% of ceVB while delivering a 1.2 dB improvement. In the figures, ceVB is omitted from Fig. 4(a), and deEP is excluded from Fig. 4(b) due to their significant performance degradation in corresponding scenarios.

VI. CONCLUSION

This letter presents the EP-dAMP detection algorithm for CF-mMIMO systems and further proposes the LC-dAMP detector to reduce the AP-side computational complexity. Simulation results and complexity analysis demonstrate that the proposed algorithms outperform existing detectors for CF-mMIMO while maintaining relatively low complexity.

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